

SMART INDIA HACKATHON 2026

TITLE PAGE

PashuPoshan AI: Smart AI-Enabled Rapid Feed & Silage Quality Testing System

- **Theme:** Agriculture, FoodTech & Rural Development
- **PS Category:** Software (Problem Statement ID: **26111**)
- **Ministry:** Ministry of Fisheries, Animal Husbandry & Dairying (DAHD)
- **Team Name:** [Your Team Name]
- **Team Leader Name:** [Team Leader Name]
- **Team Members Name:** [Member 1, Member 2, Member 3, Member 4, Member 5]

CORE PROJECT HIGHLIGHTS

- ✓ **Dual-Mode Testing:** Camera Vision + Chemical Colorimetric Strips
- ✓ **Silage Scoring:** Flieg's Fermentation Index (0–100) + pH Monitoring
- ✓ **Adulteration Alert:** Rapid screening for Urea (NPN), Sand/Silica & Aflatoxins
- ✓ **Precision Rationing:** ICAR 4-Slot Total Mixed Ration (TMR) Balancer
- ✓ **100% Offline PWA:** Operates inside rural cowsheds with IndexedDB storage
- ✓ **Vernacular & Audio:** 5 Indian Languages (Hindi, Marathi, Gujarati, Punjabi, English)

IDEA TITLE: PASHUPOSHAN AI

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed Explanation of the Proposed Solution:

- **PashuPoshan AI** is an offline-first, mobile web application (PWA) that empowers smallholder dairy farmers to rapidly test cattle feed and silage quality right at the cowshed gate.
 - **Dual-Mode Rapid Sensing:** Combines smartphone camera pixel analysis (surface texture variance, dark mold clustering) with rapid **colorimetric test-strip assays** (Euclidean color distance ΔE to quantify synthetic urea and silage pH).
 - **Closed-Loop ICAR Ration Balancer:** Instead of evaluating feed in isolation, the tested sample automatically populates its matching slot in an **ICAR (2013) 4-slot composite ration** (Green Fodder + Dry Straw + Concentrate + Minerals).
- ### • How it Addresses the Problem:
- **Eliminates Lab Lag & High Cost:** Replaces 10–14 day district laboratory turnaround and ₹1,500/sample fee with instant (<3 seconds) on-farm screening at ₹0 hardware cost.
 - **Halts Fatal Adulteration:** Directly identifies compound cattle feed spiked with toxic urea (NPN > 0.5%) and insoluble silica sand before feeding, preventing acute rumen bloat.
 - **Prevents Silage Spoiling & Aflatoxin M1:** Tracks fermentation days, flags aerobic heating (>38°C), and grades fermentation using Flieg's index to stop mould contamination in milk.

• Innovation and Uniqueness of the Solution:

1. **Flieg's Silage Index Engine:** Automated mathematical scoring ($\$220 + [2 \times \%DM - 15] - 40 \times \text{pH}$) classifying silage into Excellent, Good, Fair, or Spoiled tiers.
2. **Strict Regulatory Grading:** Correctly flags a 19.8% Crude Protein concentrate as *Tier B: Sub-Standard* failing the legal 20.0% minimum under **BIS IS:2052 Table 1** (no rounding up).
3. **Clinical Safety Triage:** Replaces dangerous DIY chemical drenches with immediate batch segregation protocols and a **1-tap call to the National Animal Disease Helpline (1962)**.
4. **Offline-First Architecture:** Uses IndexedDB for compressed image storage (avoiding 5MB LocalStorage crashes) and Service Worker v2 runtime caching for rural cowsheds.
5. **Vernacular Text-to-Speech:** Speaks diagnostic results aloud in 5 regional dairy languages (Hindi, Marathi, Gujarati, Punjabi, English) for farmers with limited literacy.

TECHNICAL APPROACH

• Technologies to be Used (Languages, Frameworks, Hardware):

- **Frontend Framework:** React 18 with TypeScript for complete type safety (zero `any` casts) and Tailwind CSS for responsive mobile UI (WCAG 2.1 touch targets $\geq 44\text{px}$).
- **Edge Sensing & Image Processing:** HTML5 Canvas API running client-side on mobile devices to extract RGB channels, luminance, texture variance, and Euclidean colorimetric distance (ΔE).
- **Storage & PWA Engine:** Native browser **IndexedDB** (`pashuposhan\_images\_db`) for high-capacity compressed image blobs + LocalStorage fallback + **Service Worker v2** Cache-First runtime caching.
- **Testing & Verification:** Vitest automated test runner with 10 passing unit tests covering Flieg math, BIS IS:2052 compliance, ICAR DMI formulas, and heuristic disclaimers.
- **Hardware Barrier:** **■0 Additional Hardware.** Runs directly inside Chrome/Safari on existing farmer smartphones without requiring expensive proprietary spectrometers.

• Methodology and Process for Implementation (Working Prototype Pipeline):

- 1. On-Farm Capture:** Farmer points camera at feed bag or dips a colorimetric strip into silage squeeze juice / feed extract.
- 2. Canvas Pixel Analysis:** Image is rendered offscreen; pixel luminance is computed. Mold spots (<60 luminance) are clustered; strip color is mapped against calibrated endpoints (Yellow=0% Urea, Magenta= $\Rightarrow$ 2% Spiked Urea, Orange=pH 3.9, Teal=pH 5.6).
- 3. Compliance & Scoring Engine:** Evaluates observed values against **BIS IS:2052:2009** (Compound Cattle Feed) and **Flieg's Index** (Silage Fermentation Quality).
- 4. Precision TMR Balancer:** Tested feed dynamically populates the appropriate slot in an **ICAR (2013)** 4-slot model, calculating composite Dry Matter Intake (DMI), Protein (g), and TDN energy.
- 5. Actionable Advisory & Export:** Delivers 1-tap call to 1962 Veterinary Emergency Helpline, TTS audio readout in 5 languages, and printable A4 PDF laboratory screening certificates.

FEASIBILITY AND VIABILITY

• Analysis of the Feasibility of the Idea:

- **Technical Feasibility:** 100% client-side execution means zero server bandwidth costs, zero cloud API latency, and instant operation even on entry-level Android devices (tested on Android 10+).
- **Operational Feasibility:** Operates completely offline inside rural cowsheds without cellular connectivity via Service Worker asset caching and local IndexedDB database persistence.
- **Economic Feasibility:** ■0 capital cost for farmers. Chemical test strips cost ■3–5 each, compared to ■1,500+ for lab wet chemistry or ■2.5 Lakhs for portable NIR spectrometers.
- **Institutional Feasibility:** Integrates smoothly with Dairy Cooperative Societies (DCS) like AMUL, NDDDB, and KMF as an on-farm screening layer prior to bulk chilling center intake.

• Potential Challenges & Risks:

1. **Variable Ambient Lighting:** Cowshed lighting varies from direct sunlight to dark mangers, impacting raw optical camera color readings.
2. **Browser Storage Constraints:** High-resolution photos quickly blow past browser LocalStorage 5MB quotas, risking application crashes.
3. **Clinical Liability & Safety:** Misinterpreting screening outputs as certified laboratory diagnostics could delay veterinary intervention.

• Strategies for Overcoming These Challenges:

1. **Colorimetric Paper Normalization:** Chemical test strips provide a standardized biochemical reaction, using Euclidean color space distance (ΔE) to minimize lighting sensitivity.
2. **Canvas Downsampling & IndexedDB:** Automatically compresses photos (max 400x300 JPEG 0.65) and stores blobs in IndexedDB (`pashuposhan\_images\_db`), eliminating LocalStorage crashes.
3. **Statutory Disclaimers & Emergency Helpline:** Permanently badges scans with 'Prototype Heuristic — No Laboratory Measurement' and puts critical actions behind the **1962 Emergency Hotline**.

IMPACT AND BENEFITS

• Potential Impact on the Target Audience:

- **Smallholder Dairy Farmers (80+ Million Households):** Empowers farmers with instantaneous bargaining power at the feed dealer; detects fraudulent low-protein feeds before payment.
- **Dairy Cooperatives (NDDB / AMUL / State Milk Unions):** Prevents Aflatoxin M1 and heavy adulterants from contaminating bulk milk chilling tankers, safeguarding milk purity.
- **Veterinary & Livestock Extension Officers:** Provides an instant diagnostic screening triage tool during field visits to trace community outbreaks of rumen bloat or feed rejection.

• Key Quantified Performance Metrics:

- **Turnaround Time:** Reduced from 14 days to **< 3 Seconds**.
- **Testing Cost:** Reduced from **■1,500/sample** to **■0 (Hardware-Free)**.
- **Daily Milk Yield:** Potential **+10% to +15% boost** via ICAR balanced TMR.

• Benefits of the Solution (Social, Economic, Environmental):

- Economic Benefits:

- Increases daily milk yield by **+1.5 to 2.0 Liters per cow/day** through optimal dietary Dry Matter Intake and crude protein balancing.
- Reduces cost of milk production by **■1.80 to ■2.50 per liter** by preventing expensive concentrate overfeeding and utilizing quality silage.
- Saves **■35,000+ per cow** in avoided emergency veterinary treatments and replacement costs caused by toxic urea poisoning.

- Social & Public Health Benefits:

- Halts the carryover of carcinogenic **Aflatoxin M1 into fluid milk**, protecting the health of children and consumer families.
- Eliminates literacy barriers with complete voice audio guidance in 5 vernacular languages.

- Environmental Benefits:

- ICAR precision ration balancing minimizes excess nitrogen excretion in dung and mitigates enteric methane emissions per kg of milk produced by **12%–15%**.

RESEARCH AND REFERENCES

• Details / Links of the Reference and Research Work:

1. **Bureau of Indian Standards (BIS):** *IS:2052:2009 (Reaffirmed 2020) — Compounded Feeds for Cattle Specification*. Table 1 prescribes minimum 20.0% Crude Protein for Type II compound feed, maximum 11.0% moisture, and maximum 3.5% acid-insoluble ash. <https://standardsbis.bsbedge.com>
2. **Indian Council of Agricultural Research (ICAR):** *Nutrient Requirements of Cattle and Buffalo (3rd Edition, 2013)*. Indian Council of Agricultural Research, New Delhi. Authoritative equations for Dry Matter Intake (DMI = 2.5% BW for indigenous zebu vs. 3.0% for crossbreds), crude protein maintenance (0.8g CP/kg BW), and lactation requirements. <https://icar.org.in>
3. **Food Safety and Standards Authority of India (FSSAI):** *Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011*. Establishes the statutory maximum ceiling for Aflatoxin M1 in fluid milk at 0.5 µg/kg to prevent human toxicity. <https://fssai.gov.in>
4. **Kung, L., Shaver, R. D., Grant, R. J., & Schmidt, R. J. (2018):** *Silage review: Interpretation of chemical, microbial, and organoleptic components of silages*. Journal of Dairy Science, 101(5), 4020–4033. Gold-standard benchmark on fermentation acids, lactic-to-butyric ratios, and clostridial heating spoilage.
5. **K■■■ç, A. (1986):** *Silage Feed: Production, Processing and Evaluation*. Bilgehan Press, Izmir, 327 p. Mathematical formulation of Flieg's Fermentation Index: $\text{Score} = 220 + (2 \times \%DM - 15) - 40 \times pH$.
6. **Garg, M. R., Sherasia, P. L., Bhanderi, B. M., et al. (2013):** *Effects of feeding balanced rations on milk production, methane emission, and economics in dairy animals in India*. Animal Production Science, 53(9), 923–936. Empirical study by NDDB demonstrating 10–15% yield gain and enteric methane reduction via balanced rations in Indian villages.
7. **Shen, L., Hagen, J. A., & Papautsky, I. (2012):** *Point-of-care colorimetric detection with a smartphone*. Lab on a Chip, 12(21), 4240–4243. Foundational paper on smartphone colorimetric detection using Euclidean color distance (ΔE) in color space.